

MAT 426, Project: Image Compression using SVD

(Due Tuesday, May 12, 2026)

You may work individually or in a group of two. You will need to work on a computer using either Matlab or Octave. The files is too large for Octave online to handle with a free account. You need to answer the questions below, provide a printout of your Matlab/Octave file, and print the figures that you generate. You may not use any outside resources, including but not limited to generative AI, internet resources, or textbooks. You are welcome to ask me any questions. Please turn in one copy per group, neatly written, well-organized, and stapled.

Definition. Let A be an $m \times n$ matrix. A matrix norm $\|A\|$ is a number assigned to the matrix A such that

- (1) $\|A\| \geq 0$ and $\|A\| = 0$ if and only if A is the zero matrix
- (2) $\|\alpha A\| = |\alpha| \|A\|$ for any scalar α
- (3) $\|A + B\| \leq \|A\| + \|B\|$

Definition. The Frobenius norm for a matrix A is defined by $\|A\|_F = \left[\sum_{j=1}^n \sum_{i=1}^m |a_{ij}|^2 \right]^{1/2}$

1. Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ -3 & 4 & 0 \end{bmatrix}$. Compute $\|A\|_F$.
2. Show that $\|I_n\|_F = \sqrt{n}$, where I_n is the $n \times n$ identity matrix.
3. Show that $\|A\|_F$ is a matrix norm
4. Show that $\|A\|_F^2 = \text{trace}(A^T A)$ where $\text{trace}(A)$ is the sum of the diagonal entries of A .
5. (a) Show that $\text{trace}(AB) = \text{trace}(BA)$.
(b) Use (a) to show that $\|AO\|_F = \|A\|_F$ for any orthogonal matrix O .
(c) Use (b) to show that $\|A\|_F^2 = \sigma_1^2 + \sigma_2^2 + \dots + \sigma_r^2$. (*Hint: $A^T A$ is a symmetric matrix.*)
6. (a) Prove that if the columns of B are linearly dependent, then so are the columns of AB .
(b) Prove that $\text{rank}(AB) \leq \text{rank}(B)$.
(c) Prove that if U is invertible, then $\text{rank}(UA) = \text{rank}(A)$.
(d) Prove that if V is invertible, then $\text{rank}(AV) = \text{rank}(A)$.
7. (a) Let $A = U\Sigma V^T$ be a singular value decomposition of an $m \times n$ matrix A . Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$ be all the nonzero singular values of A . Prove that the rank of A is r .
(b) Let $A = U\Sigma V^T$ be a singular value decomposition of an $m \times n$ matrix A . Let $k \leq r = \text{rank}(A)$. Define $A_k = U\Sigma_k V^T$, where $\Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0 \dots 0)_{m \times n}$ where $\sigma_1 \geq \sigma_2 \geq \dots \sigma_k > 0$. Prove that the rank of A_k is k . (Note: It is possible to prove that $A_k = U\Sigma_k V^T$ is the best rank(k) approximation to A , but this is beyond the scope of this project.)

8. The following questions consider how the approximation in Exercise 7 is helpful in term of image compression. Consider the outer product form

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$$

- (a) Suppose we have a grayscale picture that is 3072×2049 pixels in size. Each picture is one of 256 shades of gray represented by a number between 0 and 255. Assuming this information is stored in a matrix and we wanted to send it to someone digitally, how many numbers would have to be transmitted to share all of the information?
- (b) To reduce the amount of information that is sent, suppose we instead sent a rank 200 approximation. What information would have to be sent and how many numbers would need to be sent?
- (c) Why do the smallest singular values contribute very little to the image?
- (d) Lets see this in action using either Octave or Matlab. In Moodle, there is an image saved as `EiffelTower.jpg` that you will consider. To load this into Matlab or Octave and convert it to gray scale type

```
Eiffel=im2double(rgb2gray(imread('EiffelTower.jpg')));
```

Next, compute the singular value decomposition of the image using

```
[U,S,V] = svd(Eiffel);
```

- i. Make a plot of the magnitude of the singular values using a log scale as well as a plot of the cumulative percent of total sigmas. Save and print these images.

```
sigmas = diag(S);  
figure; plot(log10(sigmas)); title('Singular Values (Log10 Scale)');  
figure; plot(cumsum(sigmas)/sum(sigmas));  
title('Cumulative Percent of Total Sigmas');
```

- ii. Based on these images, about how many singular values should be kept to retain approximately 75% of the information contained in the Eiffel Tower?
- iii. Finally, lets make some low rank approximations to the Eiffel Tower and see how the images turn out. Save and print these images.

```
figure; imshow(Eiffel), title('Full-Rank Eiffel Tower');  
ranks = [200, 150, 100, 50, 30, 20, 10];  
for i = 1:length(ranks)  
    approx_sigmas = sigmas;  
    approx_sigmas(ranks(i):end) = 0;  
    ns = length(sigmas);  
    approx_S = S;  
    approx_S(1:ns, 1:ns) = diag(approx_sigmas);  
    approx_Eiffel = U * approx_S * V';  
    figure; imshow(approx_Eiffel),  
        title(sprintf('Rank %d Eiffel Tower', ranks(i)));  
end
```